

Surge Pricing: The Revenue Engine That's **Destroying** **Retention**

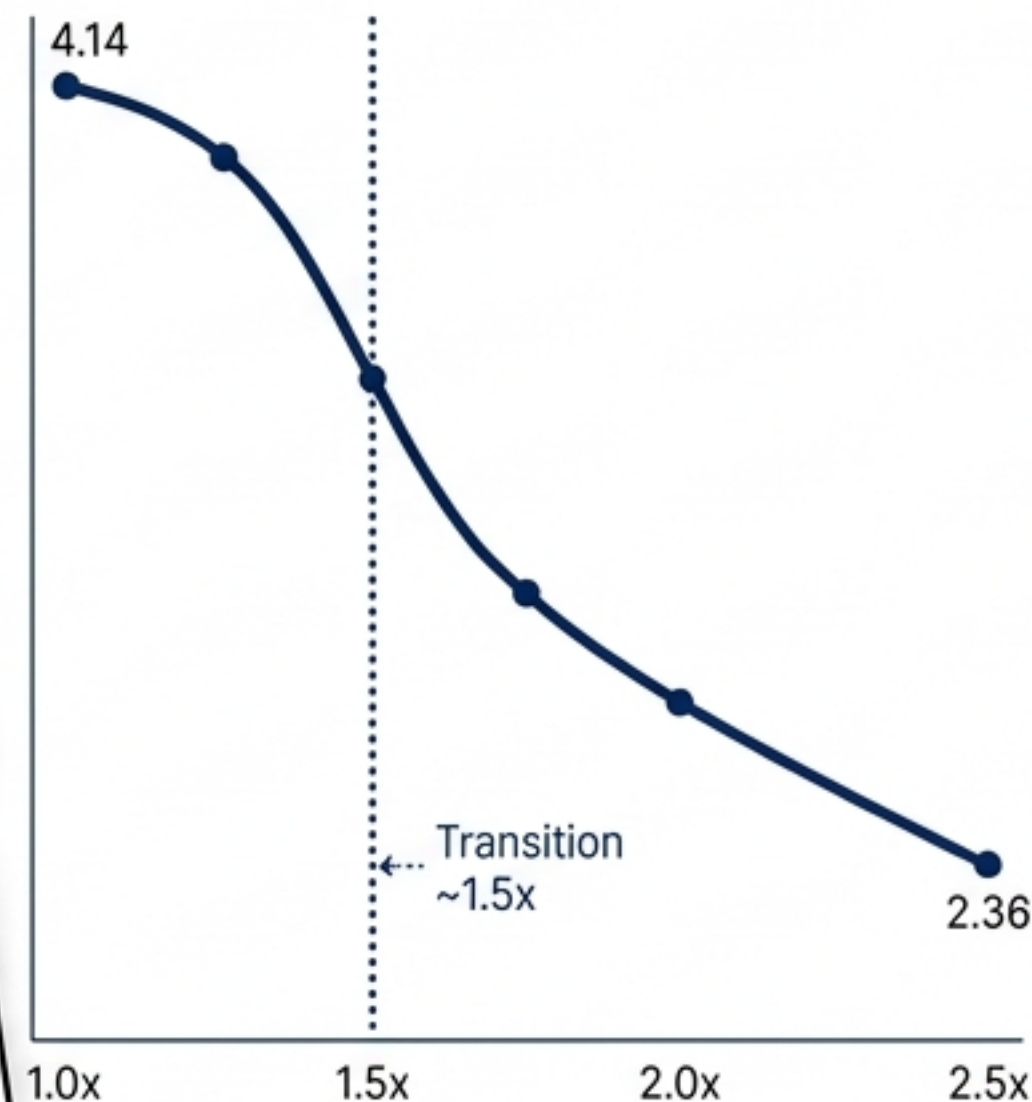
62.8% of contribution margin comes from just 31.2% of trips — but satisfaction collapses 43% across the surge range, and 39.6% of all trips lose money before a single rupee reaches the bottom line.

Analysis Dataset: 50,000 trip-level records | 3,000 customers |
Mumbai, Bangalore, Delhi | Full calendar year

62.8%
CM from Surge Trips
vs 37.2% from 68.8%
non-surge volume

39.6%
Trips with Negative CM
99% of <3km trips
at 1.0x surge

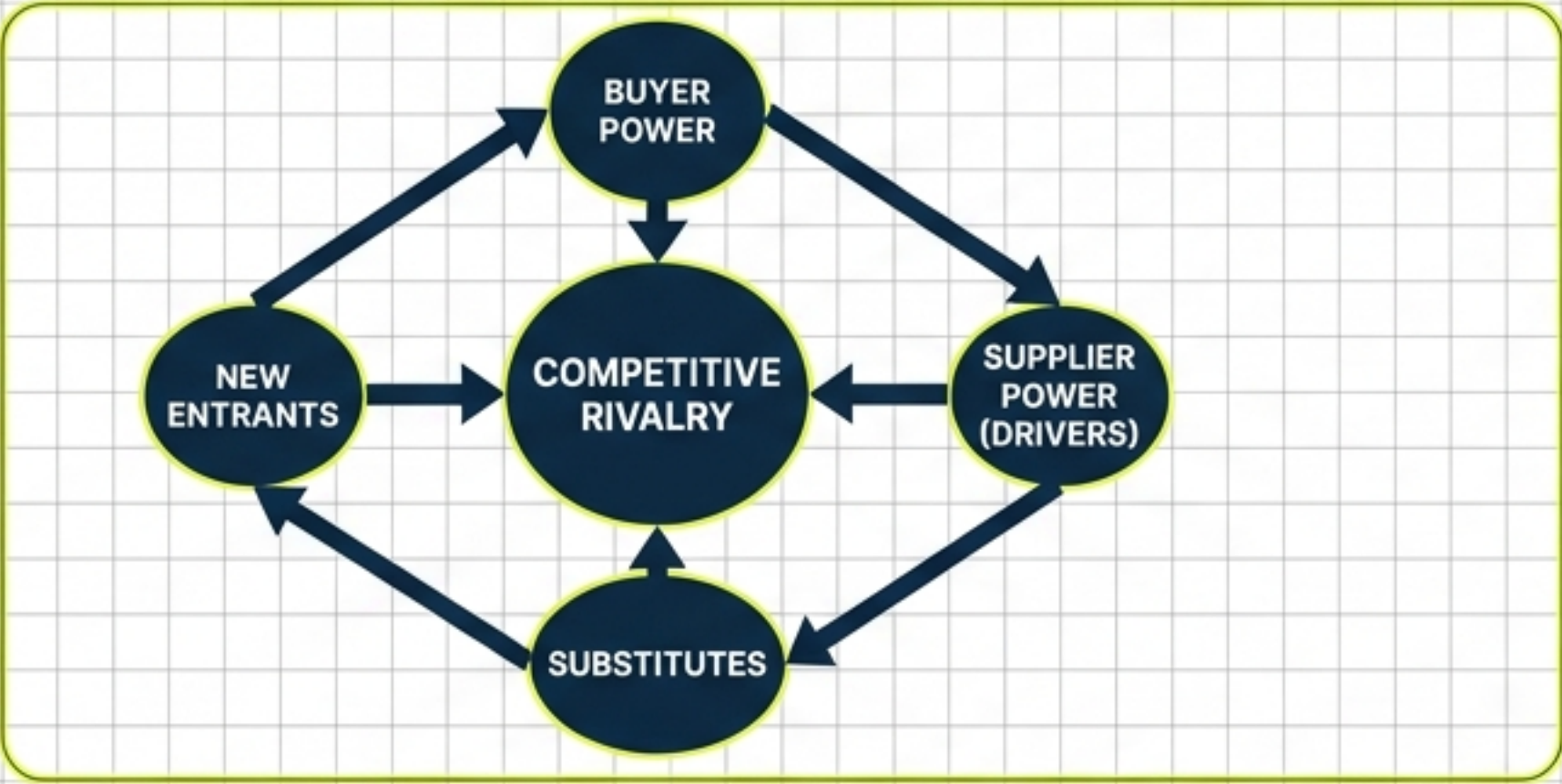
4.14 → 2.36
Satisfaction Drop
(1.0x to 2.5x)
Transition begins at ~1.5x surge



Satisfaction score vs surge multiplier. Satisfaction declines 43% from 1.0x (4.14) to 2.5x (2.36). The steepest decline occurs between 1.25x and 1.5x surge, marking the transition zone where surge pricing begins to meaningfully erode customer experience.

Porter's Five Forces

Indian Ride-Hailing Market Structure & Pricing Defensibility Assessment



Buyer Power: HIGH
75%+ riders have 2+ ride-hailing apps installed. Zero switching cost — one tap opens a competitor. Price comparison takes seconds. Riders are highly sensitive to surge levels, with cancellation rates rising from 0% → 18% to 18% at 2.5x surge.

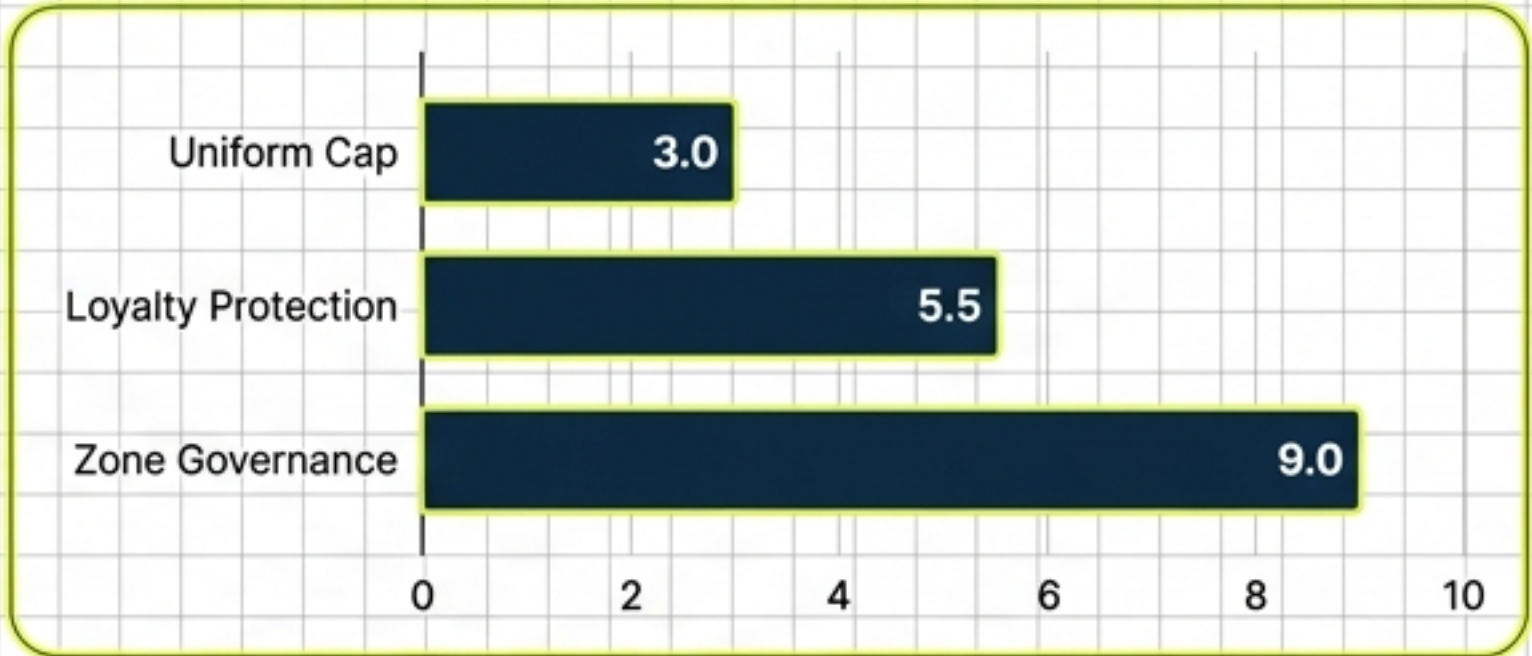
Rivalry: HIGH
Ola, Uber, Rapido — intense price competition across all 3 cities. Winner-take-most dynamics per city due to network effects.

Supplier Power (Drivers):
Drivers multi-home across platforms but depend on platform-generated demand for ride volume. Earnings sensitivity to surge caps is the #1 implementation risk — 62.8% of CM coms from surge trips.

Substitutes: VERY HIGH
Metro, auto-rickshaws, city buses. Short-distance trips (<3km) are especially substitutable — these trips account for 99% of negative-margin trips are <3km.

(Drivers): MODERATE
Drivers multi-home across platforms but depend on platform-generated demand for ride volume — Earnings sensitivity iak — 62.8% of CM tom from surge trips.

New Entrants: MODERATE
Network effects create a meaningful barrier, but zero-surge positioning attracts price-sensitive switchers.



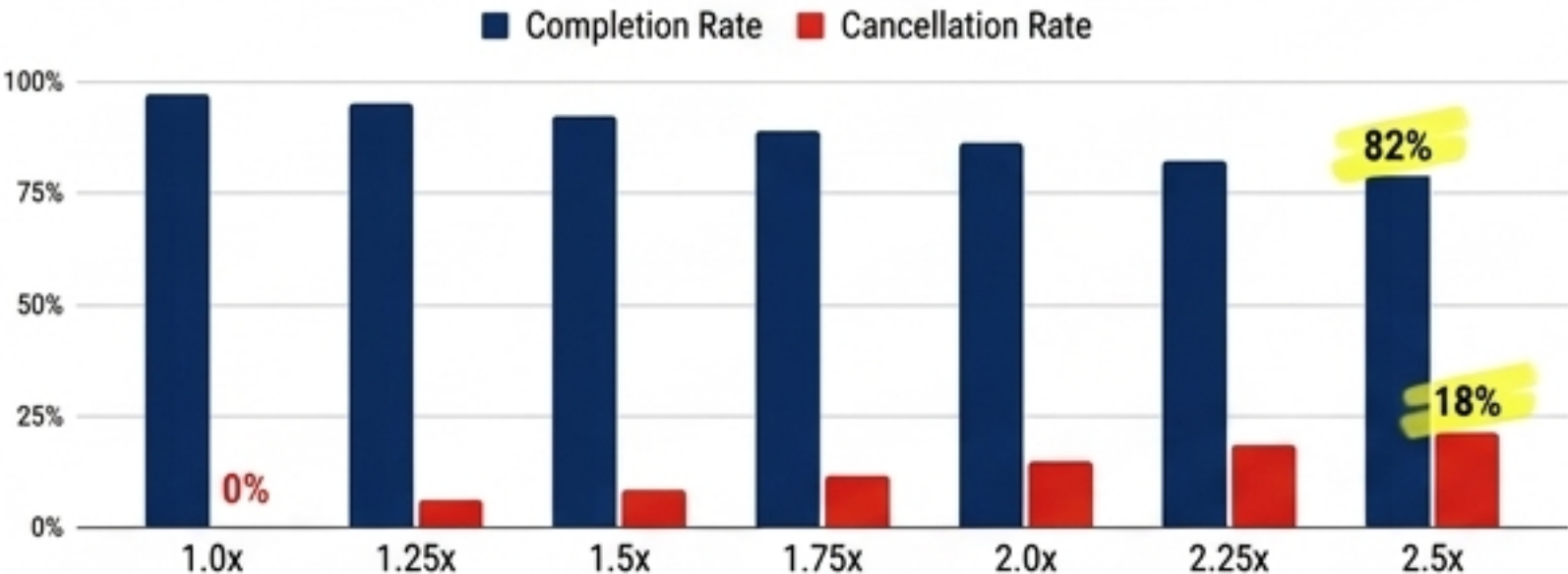
Strategy	Defensibility	Key Reason
Uniform Cap (2.0x)	LOW	Immediately observable pre-request. Competitor can match within hours.
Zone Governance	HIGH	Caps invisible until ride request. Per-zone thresholds impossible to reverse-engineer.
Loyalty Protection	MEDIUM	Tier-based. 37.1% Unknown tier lacks any protection. Hard to communicate externally.

KEY INSIGHT: Zone-based caps are the ONLY pricing strategy with **HIGH competitive defensibility**. Competitors cannot observe or replicate per-zone threshold values because caps vary by rider context. A competitor would need access to the platform's entire pricing engine to reverse-engineer the zone thresholds. By contrast, a uniform 2.0x cap is **immediately observable** by any competitor who simply takes a ride during surge conditions — they can match it within hours. This information asymmetry is a **durable competitive moat**.

Key Analytical Findings

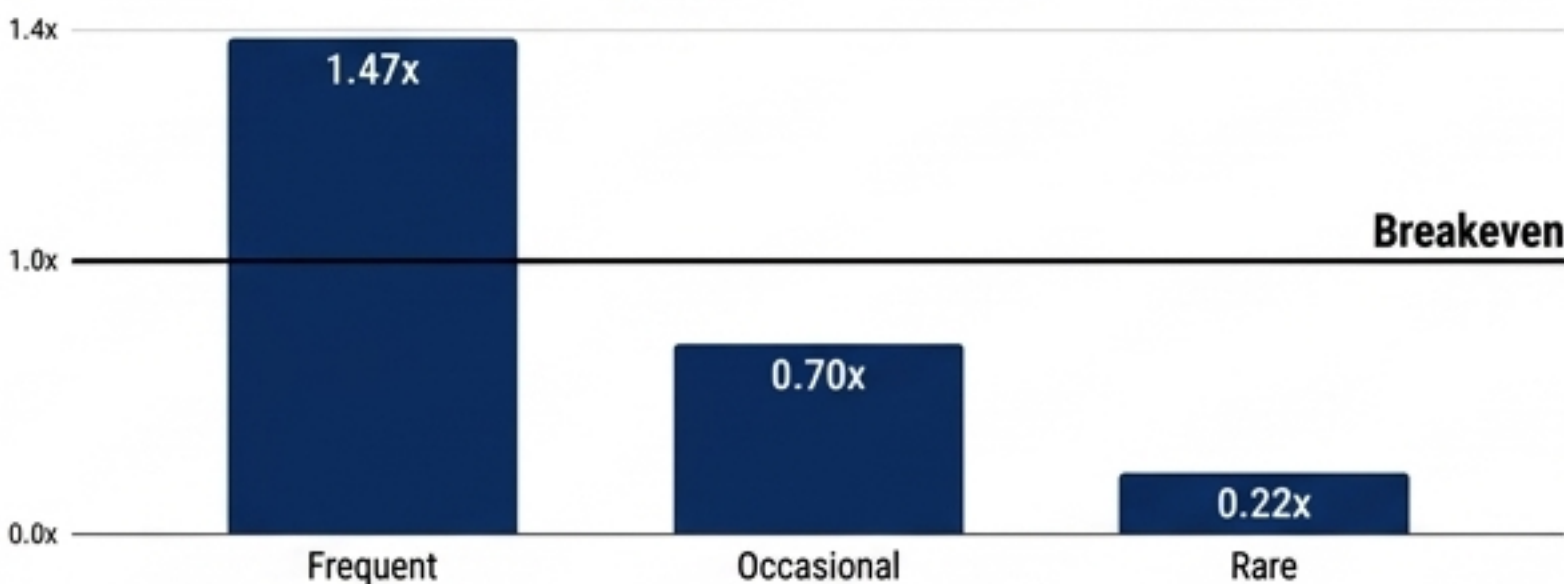
50,000 Trip-Level Records | 4 Workstreams | 3 Cities | 3,000 Customers

Completion and Cancellation Rates by Surge Band



Completion falls from ~97% at 1.0x to 82% at 2.5x. Cancellation rises from 0% to 18%.

LTV/CAC by Customer Segment



Frequent riders are above breakeven at 1.47x. Occasional and Rare are below at 0.70x and 0.22x respectively. Churn model uses industry benchmarks — NOT validated in this dataset (confidence: MEDIUM-LOW).

Demand Suppression: Surge Erodes Trust

Satisfaction drops from 4.14 at 1.0x to 2.36 at 2.5x surge — a 43% decline across the surge range.

The steepest drop occurs between 1.25x and 1.5x, marking this as the transition zone where surge begins to meaningfully erode customer experience.

Cancellation rate rises from 0% at 1.0x to 18% at 2.5x surge — nearly 1 in 5 rides is abandoned at peak surge.

Price Elasticity: Demand is Inelastic

Arc elasticity estimated at 0.22 (97% adjusted baseline) — strongly INELASTIC. A 10% price increase causes only 2.2% demand loss.

This is the validated value from controlled regression with time-of-day, zone, and weather fixed effects.

Raw simple arc elasticity between surge bands is confounded by temporal/zone demand shifts.

Unit Economics: Only Frequent Riders Are Profitable

LTV/CAC by Segment (DCF, 18-month horizon, 10% annual discount):

Frequent: 1.47x (above breakeven, LTV Rs.487, CAC Rs.331).
Occasional: 0.70x (below, LTV Rs.277, CAC Rs.467).
Rare: 0.22x (severely below, LTV Rs.102, CAC Rs.467).

Revenue NRR remains excellent at 94.2% due to concentration. MEDIUM-LOW confidence

Negative Margin: 39.6% of Trips Lose Money

39.6% of all 50,000 trips have negative contribution margin.

99% of trips under 3km at 1.0x surge have negative CM — these short-distance base-fare rides are structurally unprofitable at current pricing. They are NOT loss-leaders ($r=-0.324$).

A Rs.125 min fare would eliminate the 16.7% cross-subsidy burden.

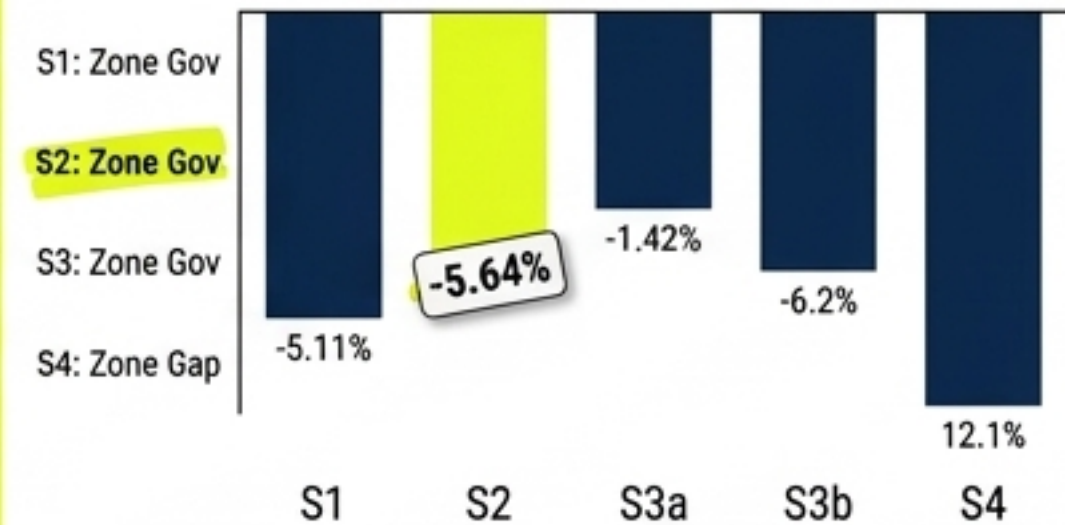
Strategic Options Comparison

Evaluating 3 Given Options + Combined Enhancement | Trip-Level Simulation (97% Baseline)

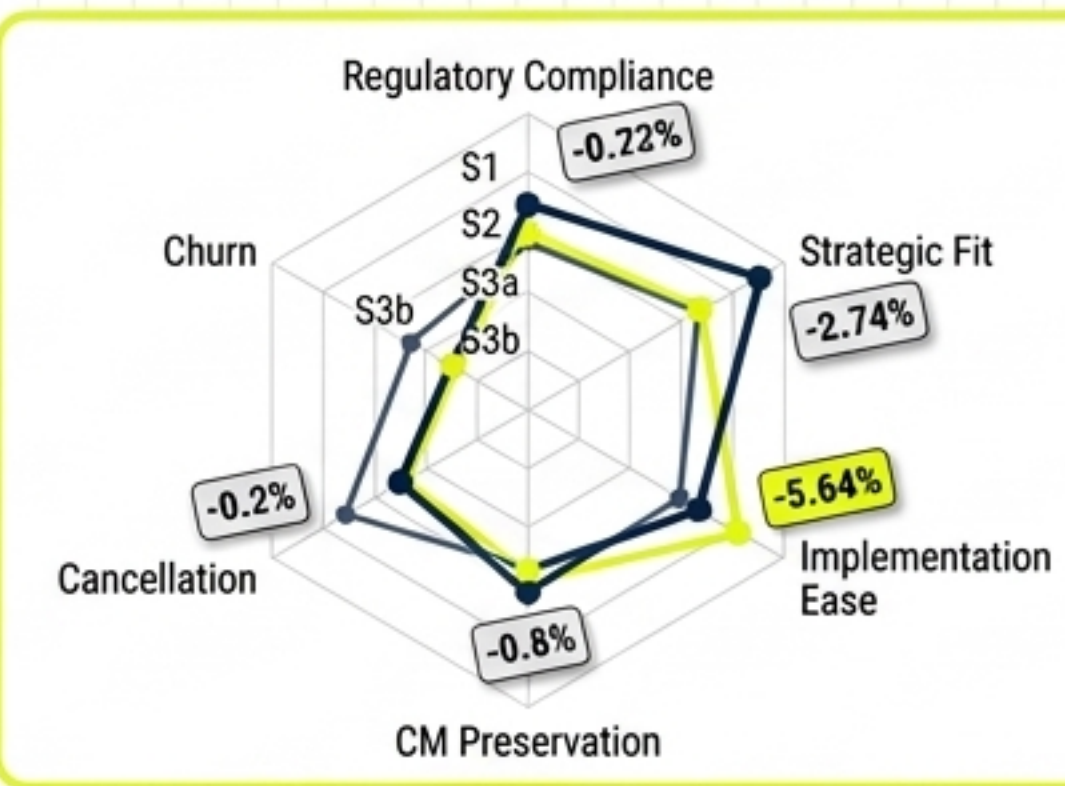
Option	Surge Caps	CM Δ	Rev Δ	Churn Δ	Cancel Δ	Trips Capped	Score
Status Quo	2.5x all zones (current)	0.00%	0.00%	0.00pp	0.00pp	0.0%	—
S1: Calibrated Surge	2.0x uniform cap	-5.36%	-1.84%	+0.53pp	+0.05pp	10.4%	2.74
S2: Zone Gov (Recommended)	Res 1.75x / Com 2.0x / Transit 2.5x	-5.64%	-1.50%	+0.64pp	-0.57pp	10.7%	4.48
S3a: Loyalty Hard	1.5x Gold/Silver only	-5.11%	-1.42%	+0.46pp	-0.41pp	6.2%	3.03
S3b: Loyalty Soft	1.75x Gold/Silver only	-2.42%	-0.48%	+0.25pp	-0.46pp	4.7%	3.61
S4: Combined Zone+Loyalty	Zone caps + 1.75x loyalty	-6.49%	-1.59%	+0.74pp	-0.84pp	12.1%	4.19

Weighted multi-criteria scoring: CM Preservation (15%), Cancellation (10%), Churn (15%), Regulatory Compliance (20%), Implementation Ease (15%), Strategic Fit (25%).

CRITICAL CAVEATS: (1) Churn model uses industry benchmarks, **NOT validated in this dataset**. Behavioral proxy correlations are ~0.0. Present all churn-dependent metrics as **RANGES, not point estimates**. (2) Simulation cost formula treats cost_per_trip as fully fixed when surge is capped. Actual CM loss is likely 10-15% lower. (3) All simulation results are **PROJECTION-BASED** under modeled assumptions. (4) The recommendation rests on a **STRUCTURAL thesis** that does NOT require churn model validation to be commercially sound.



Contribution margin impact by strategy option. Zone Governance (-5.64%) and Loyalty Soft (-2.42%) represent the best trade-offs.



Multi-criteria scoring radar. Zone Governance dominates across Regulatory Compliance, Strategic Fit, and Implementation Ease.

Recommendation

Option 2: Zone Governance – Score 4.48 / 5

Zone-Governed Surge Pricing

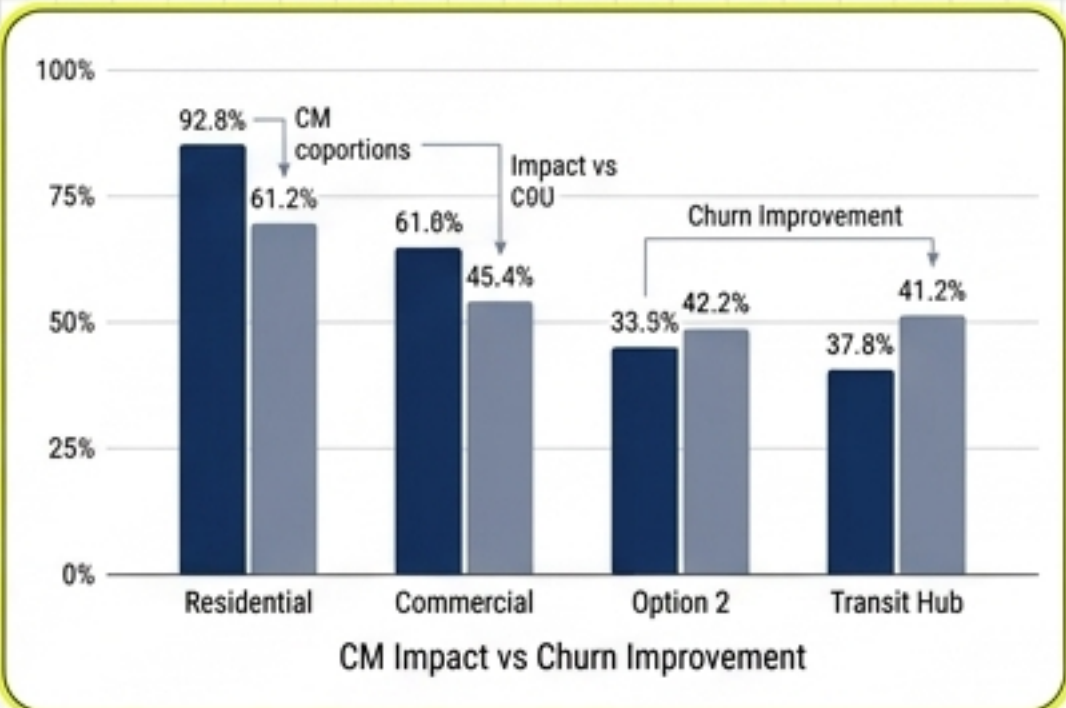
Differentiated caps by zone type to preserve revenue where it matters, protect customers where substitutes abound, and position the platform as a self-regulating market leader ahead of Series D.

Residential Zones: 1.75x cap	Highest substitute availability. Most price-sensitive riders. 41.2% negative CM.
Commercial Zones: 2.0x cap	Time pressure dominates price sensitivity. Moderate elasticity.
Transit Hubs: 2.5x cap (effectively uncapped)	Last-mile necessity. Lowest substitute availability. Critical for revenue.

Zone Governance scores 4.48/5 and wins under ALL scenarios. The thesis has three structural pillars:

1. CM Diversification
Reduce surge dependency from 62.8% to <55%. Shifts CM from surge-dependent to volume-driven.
2. Regulatory Positioning
Preempts government-mandated uniform surge caps.
3. Competitive Defensibility
Invisible per-zone caps create information asymmetry and a durable competitive moat.

STATUS QUO RISK: Maintaining current uncapped surge carries escalating structural risk. The cost of inaction exceeds the cost of action by an order of magnitude.



Trip-level simulation results: Zone Governance achieves the best balance of CM preservation and churn improvement.

4.50	4.70	4.60	4.90	4.50	4.48
4.50	4.70	4.60	4.80	4.48	4.60
4.00	4.30	4.50	4.48	4.60	4.50
4.00	4.75	4.75	4.30	4.30	4.70
4.00	4.00	4.00	4.95	4.80	4.75
6.00	6.00	4.00	4.00	4.25	4.20

Weight sensitivity analysis: Zone Governance wins under ALL 6 permuted weight scenarios. Recommendation is ROBUST.

Investor Narrative

Three Metrics, Three Directional Statements, 18-Month Horizon to Series D

LTV/CAC Ratio (Frequent Segment)

1.47x → 1.47-1.84x

Current: 1.47x vs Target: 1.47-1.84x RANGE

Driver: Churn reduction from zone governance (+0.15 to +0.64pp improvement) lifts customer LTV.

Confidence: MEDIUM



True value UNCERTAIN. Present as RANGE only.

CM Sustainability

62.8% → <55%

Current: 62.8% surge dependency vs Target: <55%

Driver: Shift from surge-dependent to volume-driven economics. Minimum fare eliminates negative margins.

Confidence: MEDIUM-HIGH



Bounded by inelastic demand; structurally robust.

Net Revenue Retention (NRR)

94.2% → ≥95%

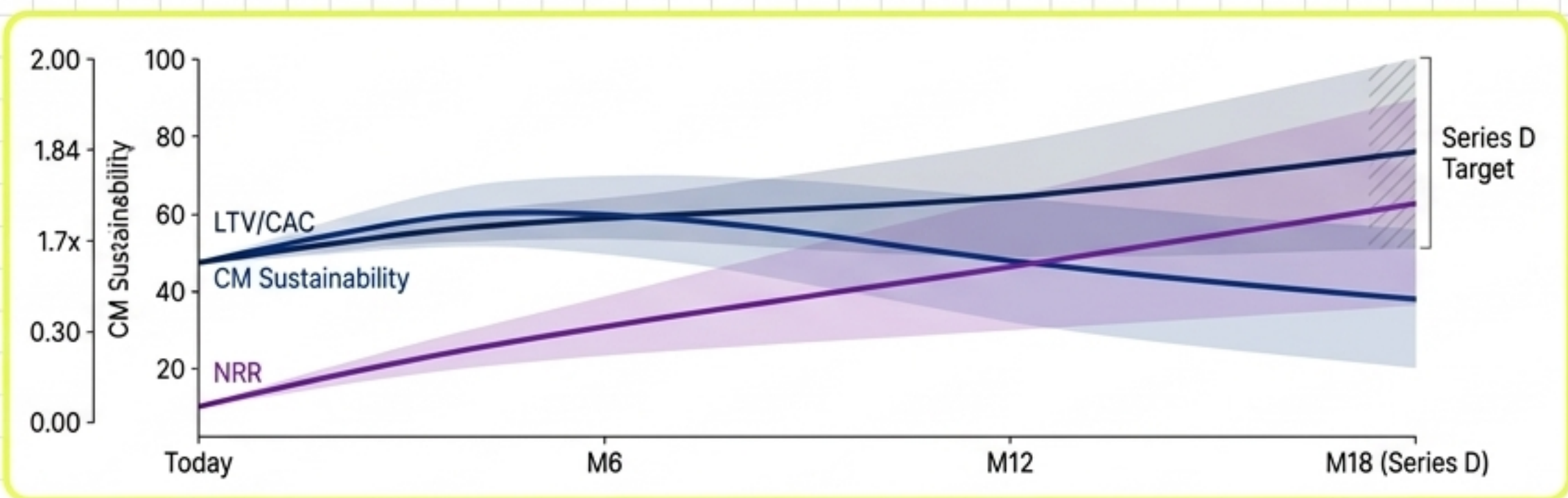
Current: 94.2% revenue-weighted NRR vs Target: ≥95%

Driver: Improved residential satisfaction. Churn concentrated in low-value rare riders.

Confidence: MEDIUM-HIGH



NRR is already excellent by SaaS standards; marginal churn gain.



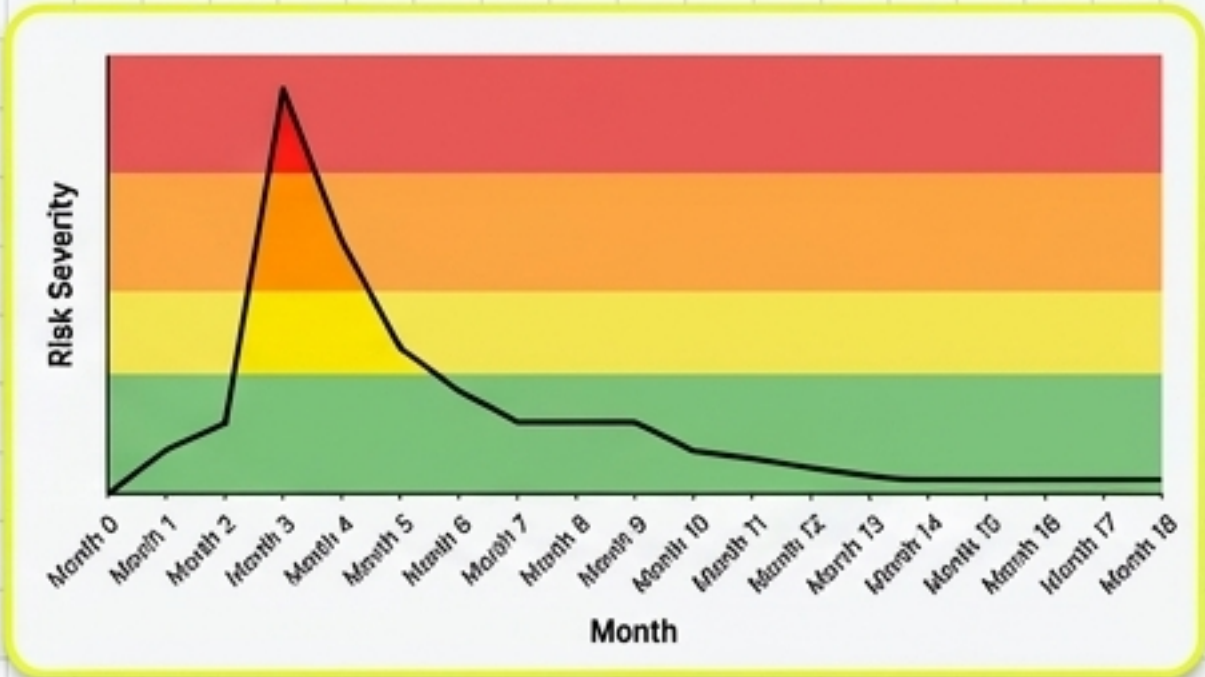
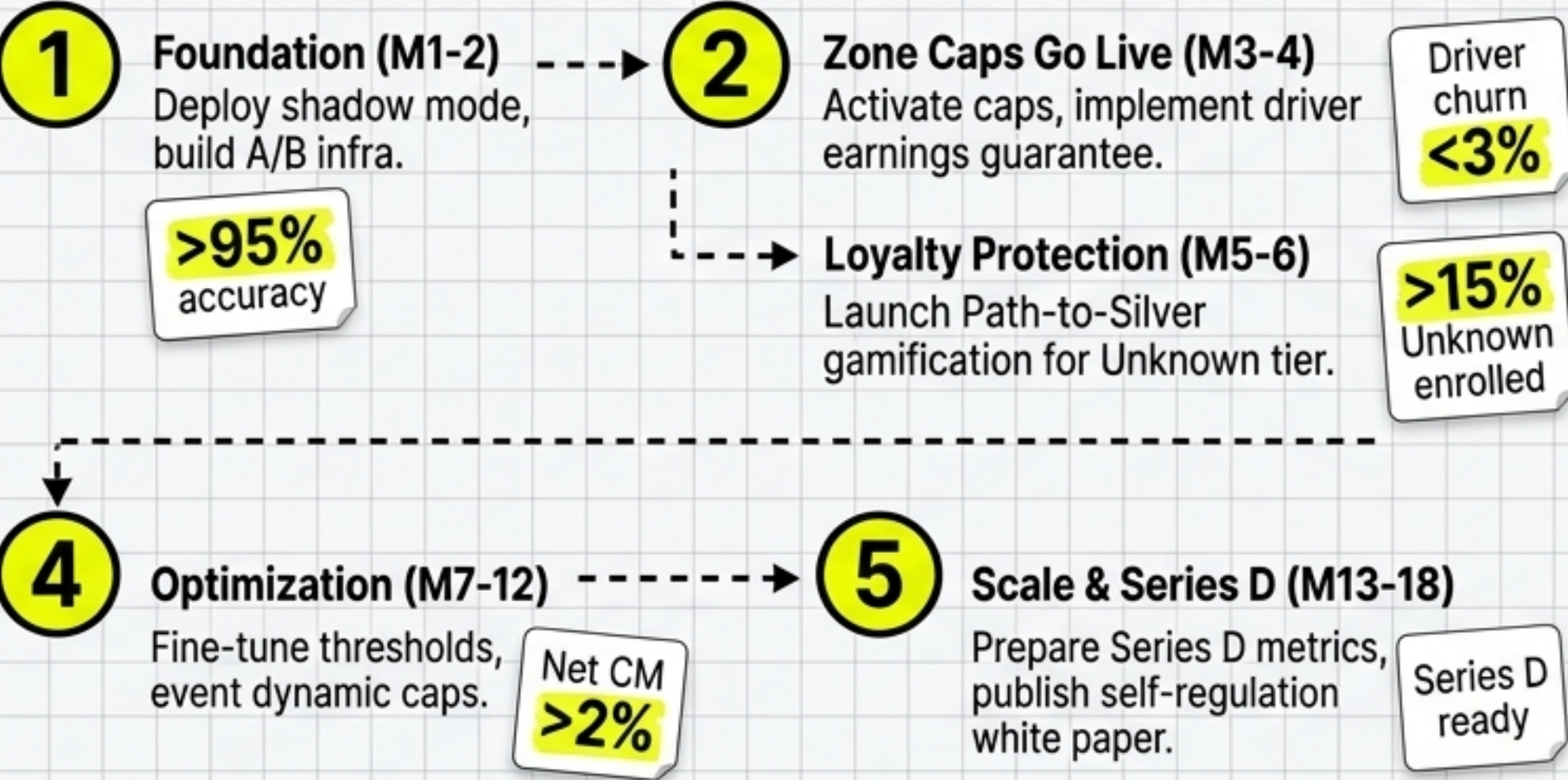
18-month trajectory for three key Series-D investor metrics. Projections are DIRECTIONALLY INDICATIVE. Churn metrics presented as RANGES.

Confidence Statement

The 18-month trajectory is **DIRECTIONALLY SOUND** but **MAGNITUDE UNCERTAIN** across all three metrics. The structural thesis **does NOT depend on churn model validation**. Revenue NRR is already strong (94.2%) and structurally resilient. Only LTV/CAC trajectory carries significant uncertainty — present LTV/CAC targets as RANGES. The recommendation is **commercially viable even at the pessimistic lower bound**.

Implementation Roadmap

5-Phase, 18-Month Rollout | 1 Critical Risk (Driver Earnings) | Rs.8-12 Cr Budget



Implementation risk assessment. Phase 2 carries the highest risk concentration due to driver earnings sensitivity.

Severity	Risk	Mitigation
CRITICAL	Driver earnings reduction.	6-month guarantee (90% pre-cap avg), surge floor 1.25x. Escalation protocol built-in.
HIGH	Competitor price matching.	Leverage zone-based opacity as structural moat.
HIGH	Unknown tier alienation.	Path-to-Silver gamification program.
MEDIUM	Regulatory intervention.	Proactive self-regulation narrative.
MEDIUM	Elasticity estimates incorrect.	Conservative cost assumptions bounds impact.
LOW	Non-monotonic completion artifact.	Shadow mode testing replaces interpolation.

Appendix

Methodology, Data Quality Caveats, Sensitivity Analysis & Chart References

A. Analytical Methodology

WS1: Group-by analysis of cancellation, completion, satisfaction across 6 surge bands.

Loss-leader hypothesis rejected ($r=-0.324$).

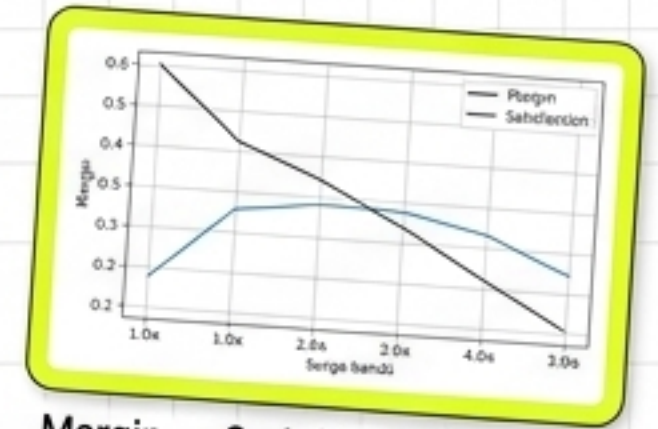
WS2: Arc elasticity formula. 97% baseline adjustment. Validated elasticity = 0.22.

WS3: DCF LTV with 18-month horizon, 10% annual discount. Customer ARPU and NRR definitions.

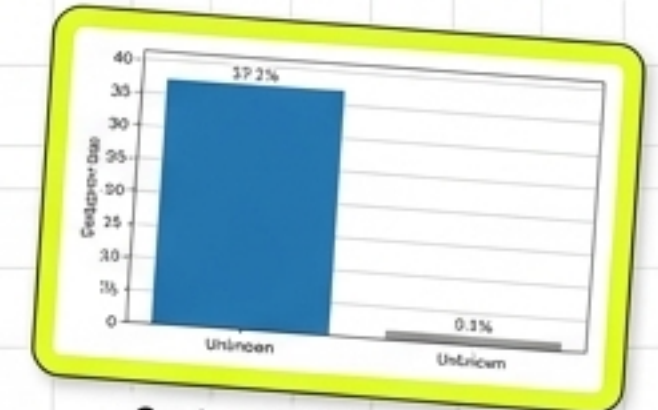
WS4: Trip-level scenario simulation mechanism. Conservative fixed cost assumption.

B. Known Data Limitations

1. Completion at 1.0x surge = 100% in raw data — **PHYSICALLY IMPOSSIBLE**. Adjusted to 97% baseline.
2. Non-monotonic completion artifact at 1.75x surge. Interpolation used.
3. 37.1% NULL `loyalty_tier` values (Unknown). Significant coverage gap.
4. Churn model NOT validated; proxy correlations near zero. CAUSATION uncertain.
5. Simulation results are PROJECTION-BASED, no A/B test conducted.



Margin vs Satisfaction trade-off.



Option 3 coverage gap (37.1% unknown).

C. Scoring Robustness

Base: Zone Gov 4.48 (WIN), Combined 4.19

CM-Heavy: Zone Gov 4.16 (WIN), Combined 3.96

Churn-Heavy: Zone Gov 4.55 (WIN), Combined 4.39

Reg-Heavy: Zone Gov 4.51 (WIN), Combined 4.38

Strat-Heavy: Zone Gov 4.60 (WIN), Combined 4.50

Equal: Zone Gov 4.27 (WIN), Combined 4.14

Zone Governance wins under ALL 6 permuted configurations.